TIC-TAC-TOE: A PYTHON PROGRAMMING PROJECT

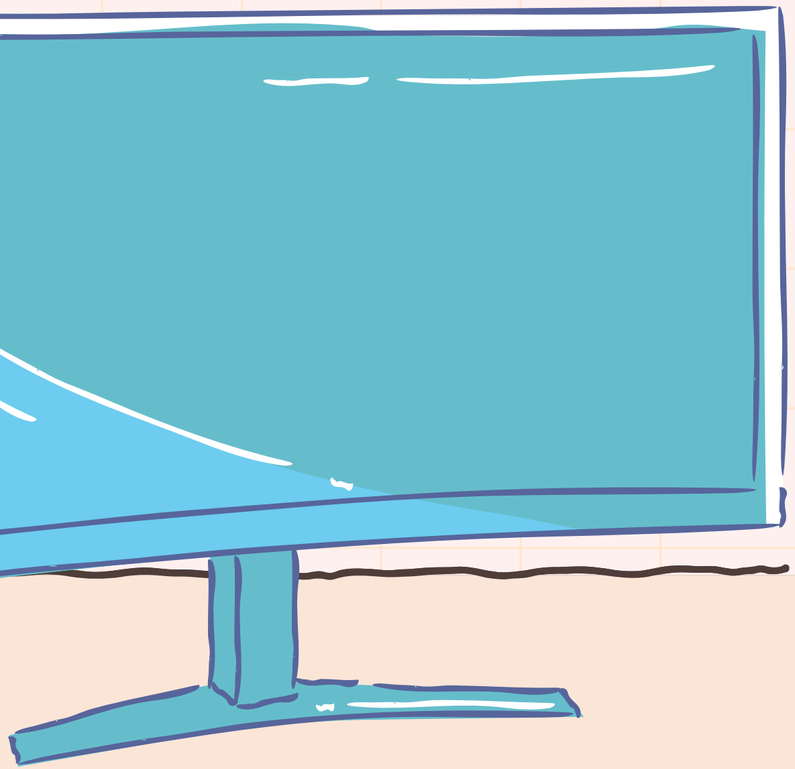
Allones, Khyle Danielle

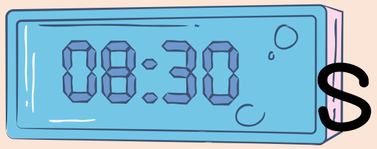
Almodovar, Michaella Jane

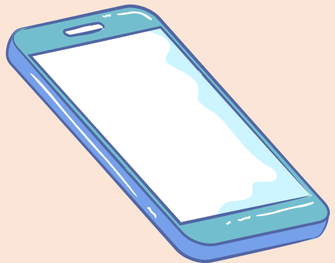
Firmo, Jhona May

Gonzales, Rainfield

Serenio, Jhoem











DESCRIPTION

The Tic-Tac-Toe game is a Python-based program designed to provide a simple and enjoyable gaming experience for two players. The game features a 3×3 grid where players take turns choosing spaces to place their symbols, X or O. The program checks each move to determine whether a player has formed a winning combination or if the game ends in a draw. Through this project, users can experience how Python can be used to create interactive programs while applying basic programming concepts and logical problem-solving skills.

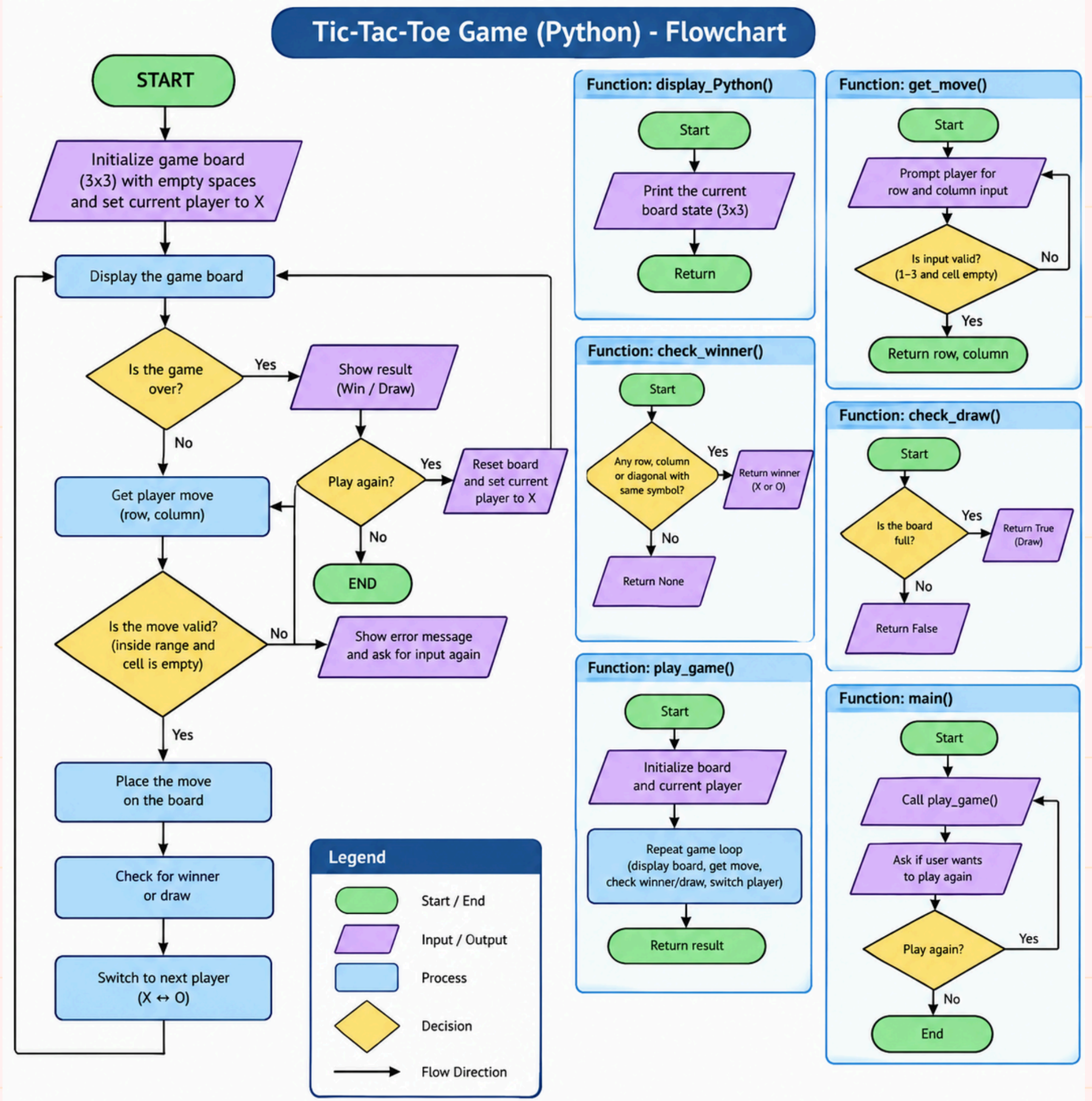
OBJECTIVES

1. To create an interactive Tic-Tac-Toe game that allows two players to compete by taking turns placing their symbols on the game board.

2. To implement game rules and validation using Python to prevent invalid moves and recognize winning combinations.

3. To improve programming and logical thinking skills by developing a game that processes player input and determines the final result.

FLOW CHART


Tic-Tac-Toe Game (Python) - Flowchart
START
Initialize game board (3x3) with empty spaces and set current player to X
Display the game board
Is the game over?
Yes
Show result (Win / Draw)
No
Play again?
Yes
Reset board and set current player to X
No
END
Get player move (row, column)
Is the move valid? (inside range and cell is empty)
No
Show error message and ask for input again
Yes
Place the move on the board
Check for winner or draw
Switch to next player (X ↔ O)
Legend
Start / End
Input / Output
Process
Decision
Flow Direction
Function: display_Python()
Start
Print the current board state (3x3)
Return
Function: get_move()
Start
Prompt player for row and column input
Is input valid? (1–3 and cell empty)
No
Yes
Return row, column
Function: check_winner()
Start
Any row, column or diagonal with same symbol?
Yes
Return winner (X or O)
No
Return None
Function: check_draw()
Start
Is the board full?
Yes
Return True (Draw)
No
Return False
Function: play_game()
Start
Initialize board and current player
Repeat game loop (display board, get move, check winner/draw, switch player)
Return result
Function: main()
Start
Call play_game()
Ask if user wants to play again
Play again?
Yes
No
End